

SAMAI TEAM

Predicting Heart Disease Risk with Machine Learning

Group members :

Simon CORBINUS – 37

Gabin CLAVIER – 36

Mathis BLANCHARD - 32

Summary :

The goal of this project is to predict if a patient has heart disease using machine learning. We will use the "Heart Disease Dataset" by *johnsmith88* from Kaggle. This dataset contains 1025 rows and 14 columns, featuring key patient details like age, sex, chest pain type, and cholesterol levels. The target variable is binary, indicating whether the patient currently has heart disease (1) or not (0).

Our workflow will cover the complete data science pipeline. First, we will clean the data using Pandas to handle any potential missing values, duplicates, or incorrect data types. Next, we will perform Exploratory Data Analysis (EDA) with Matplotlib and Seaborn to visually identify correlations between health indicators and disease risk. For the modeling phase, we will train and compare three classification algorithms: Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN).

To evaluate our models, we think that we will use accuracy, recall, and the confusion matrix. In a medical context, we will specifically prioritize reducing false negatives, as failing to identify a sick patient is dangerous.